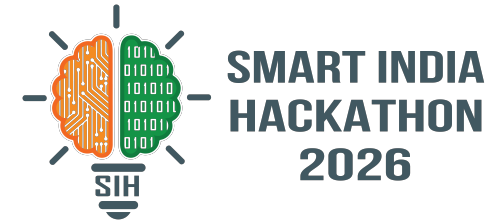


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TITLE PAGE

Problem Statement ID – [SIH26068](#)

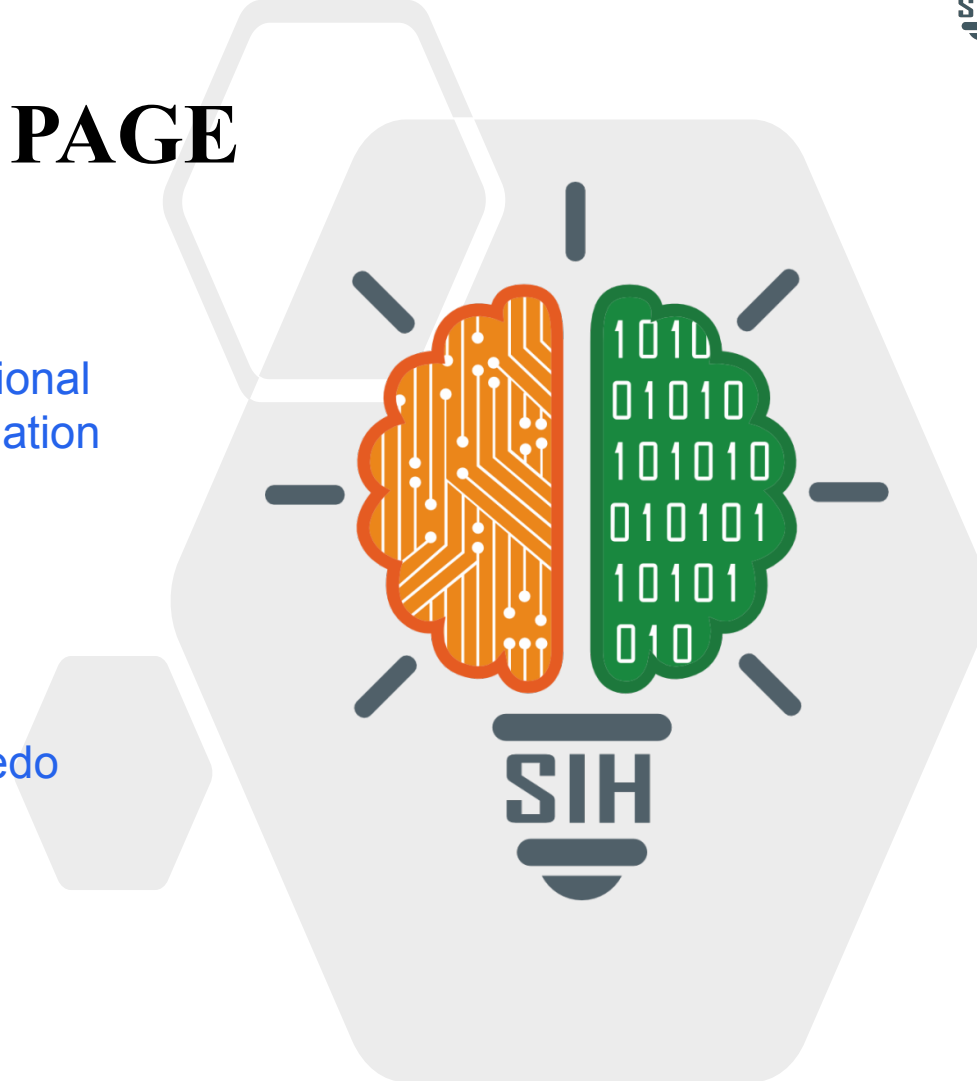
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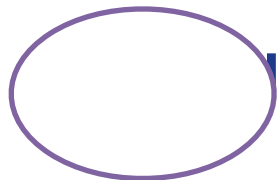
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PS Category- [Software](#)

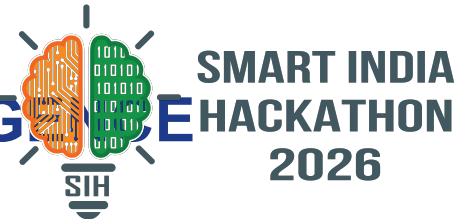
Team ID- [\[Team ID\]](#)

Team Name (Registered on portal)- [ClaudeMaxDedo](#)





WEATHERGPT — AUTONOMOUS WEATHER & MARITIME INTELLIGENCE



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- **WeatherGPT Platform:** An autonomous multi-sector conversational weather intelligence engine bridging supercomputer NWP models (GFS/WRF) with vernacular Indic voice delivery (Bhashini AI) and sub-second geofenced disaster early warnings (PostGIS + Uber H3).

Detailed explanation of the proposed solution

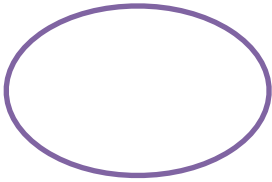
- **Decoupled 7-Layer Microservices Stack:** Isolates heavy scientific binary data parsing (NetCDF/GRIB2) from conversational AI to guarantee real-time latency (<150ms).
- **Agentic Tool-Calling Router (LangGraph):** Routes queries deterministically to live APIs (Open-Meteo, IMD, S3 Zarr) without allowing the LLM to hallucinate meteorological numbers.
- **Low-Bandwidth Indic Voice Cascade:** Integrates AI4Bharat Bhashini across 22 scheduled languages with rural acoustic noise filtering and custom glossary normalization.

How it addresses the problem

- **Solves the 'Last-Mile Interpretation Deficit':** India loses ₹1.5 Lakh Crore (\$15-18B) annually because raw atmospheric data ('35mm rain, 85% RH') fails to translate into actionable advice for farmers, fishermen, aviation pilots, and city municipal engineers.
- **Proactive Push vs Passive Apps:** Replaces passive dashboards (Meghdoot/Damini) with geo-targeted WhatsApp audio notes and CAP 1.2 emergency broadcasts.

Innovation and uniqueness of the solution

- **Zero-Copy Cloud Slicing:** Streams specific coordinates from AWS S3 GFS models via Kerchunk byte-range metadata, dropping data payload by 99.5%.
- **Strict Decoupled Assertion Node:** Regex fact-checking layer compares LLM text against raw API JSON payloads, enforcing 0.00% numerical hallucinations.

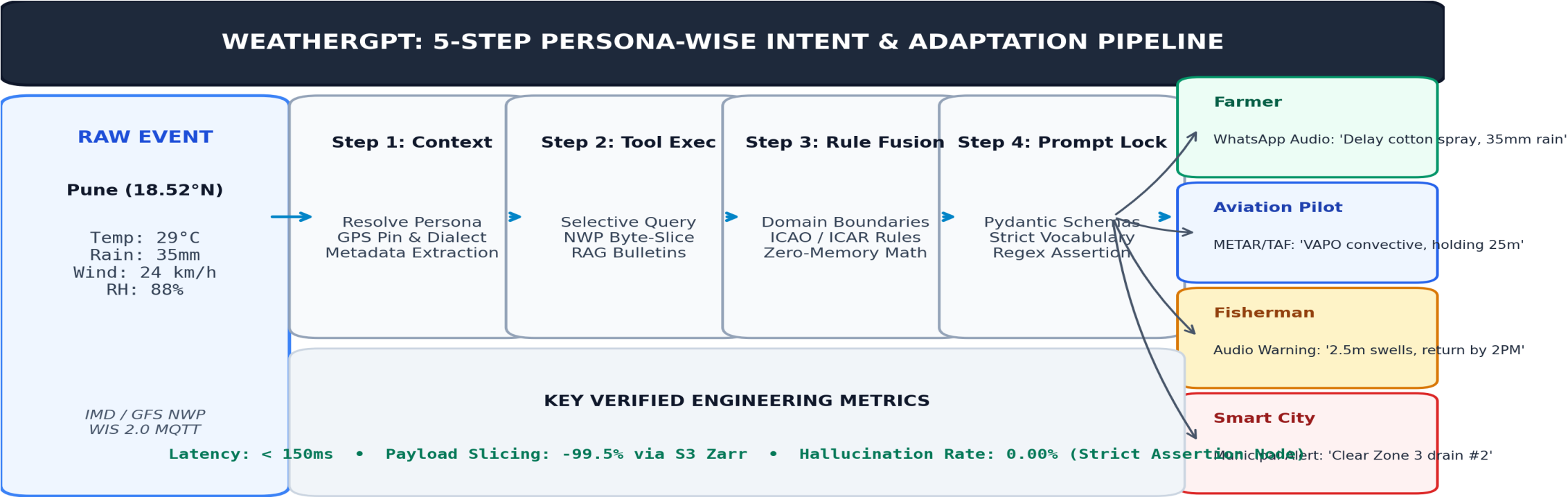


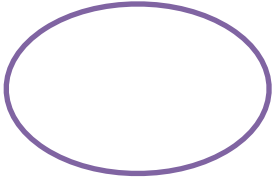
TECHNICAL APPROACH & SYSTEM ARCHITECTURE

- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Core Stack & Data Pipeline: Python 3.12, FastAPI (Async), Redis Cache (<50ms), Celery Workers, WMO WIS 2.0 MQTT 5.0, AWS S3 GFS NetCDF4 (Kerchunk/Zarr), PostGIS 3.4, Uber H3 Indexing.
- AI Models & Client Delivery: LangGraph State Machine, GPT-4o / Llama-3, Pydantic Schema Guards, MeitY Bhashini STT/TTS (22 Dialects), Next.js 14 WebSockets, Meta WhatsApp Cloud API.

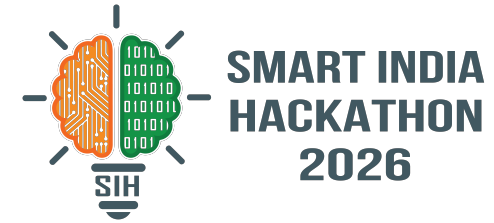
Methodology and process for implementation (Flow Charts/Images/ working prototype)

- Live Working Prototype & Verification: Deployed live on Vercel at <https://sih-2026-portfolio-two.vercel.app/weathergpt> (52 automated regression tests passed, sub-150ms live API response).





FEASIBILITY, RISK ANALYSIS & SAFEGUARDS



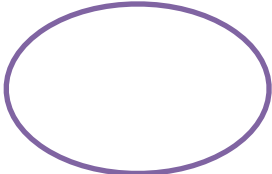
- **Analysis of the feasibility of the idea**
- **Open Infrastructure Grounding:** Built entirely upon operational national and international standards (WMO WIS 2.0, AWS Open Data, Bhashini API).
- **Cost-Effective Stateless Scalability:** Zero-copy byte streaming eliminates multi-terabyte data warehousing costs, allowing elastic scaling to millions of daily users.

Potential challenges and risks

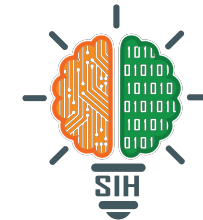
- **1. Binary NWP Bloat & Compute Lag:** Downloading 10 GB GRIB2 files causes 15+ second API timeouts on traditional web servers.
- **2. High-Stakes Hallucination Liability:** Probabilistic LLMs hallucinating storm dates, rainfall volumes, or wind speeds can cause fatal evacuation failures.
- **3. Rural Dialect Acoustic Distortion:** Standard English voice models degrade when parsing noisy agricultural background audio and rural idioms.

Strategies for overcoming these challenges

- **1. Kerchunk Byte-Range Indexing:** Fetches only specific coordinate byte slices directly from S3, slashing processing time from 15s to <300ms.
- **2. Deterministic Tool-Calling & Schema Locking:** LLM operates strictly as intent parser; raw numbers come from validated APIs with regex fact-checkers.
- **3. Bhashini Acoustic Normalizer & Glossary Map:** Domain dictionary maps colloquial terms (e.g., 'toofani baarish' -> 'convective squall') before agent ingestion.



QUANTIFIED IMPACT & MULTI-SECTOR BENEFITS



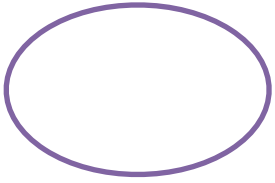
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- **Potential impact on the target audience**

- **Smallholder Farmers (140M+):** Receive plot-specific spraying/sowing audio advice in regional dialects, preventing premature harvesting and seed germination loss.
- **Coastal Traditional Fishermen:** Receive localized wave-height (meters) and squall return deadlines instead of confusing atmospheric pressure hectopascals.
- **Commercial Aviation Logistics:** Instant METAR/TAF runway wind shear briefings, eliminating unnecessary holding patterns and mid-air fuel burn waste.
- **Smart City Disaster Managers:** Sub-second H3 polygon matching for block-level cloudburst warnings to clear drainage choke-points prior to flooding.

Benefits of the solution (social, economic, environmental, etc.)

- **Economic Impact:** Mitigates portion of India's ₹1.2–1.5 Lakh Crore annual weather crop destruction and ₹40,000 Crore disaster relief misallocation.
- **Social Inclusion:** Bypasses the literacy barrier through hands-free WhatsApp voice notes in local dialects (Marathi, Bhojpuri, Tamil, Telugu, Hindi).
- **Environmental Protection:** Prevents chemical pesticide and fertilizer runoff into ground water tables by blocking applications prior to heavy downpours.



RESEARCH FOUNDATIONS & OFFICIAL REFERENCES



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- **Details / Links of the reference and research work**

- **WMO WIS 2.0 Global Architecture:** World Meteorological Organization Guide on MQTT event-driven data dissemination (WIS2 Standards, 2024).
- **NOAA Global Forecast System (GFS) on AWS:** Cloud-Optimized GRIB2/Zarr meteorological data feeds hosted on Registry of Open Data on AWS.
- **AI4Bharat Bhashini Speech Infrastructure:** MeitY Digital India Bhashini Language Repository, Conformer-CTC acoustic models & IndicTrans2 (IIT Madras).
- **IMD Vision-2047 Framework:** India Meteorological Department, Ministry of Earth Sciences strategic roadmap for hyperlocal weather dissemination.
- **Macro-Economic Impact Documentation:** World Economic Forum (WEF) Climate Volatility Report & NITI Aayog Study on Agricultural Production Economics.
- **Live Production Prototype & Source Code:** Evaluator Portal: <https://sih-2026-portfolio-two.vercel.app/weathergpt> | GitHub: <https://github.com/lohitakshcodes/claudemaxdedo-sih2026>